Danjie Tang

LinkedIn | GitHub | Google Scholar

Education

University of Toronto | *B.A.Sc. in Computer Engineering* Sept 2021 – May 2027

- **cGPA 3.88/4.00** (89.7%) | GPA by year: 3.76 → 3.95 → 3.97

Industry Experience

Apple | *Software Engineer Intern* March 2026 – August 2026

- Benchmarked aggregate and per-request generation **throughput** and **time to first token** with **vLLM** across single-node and 2-node **NVIDIA DGX Spark** configurations against **llama.cpp** on Apple Silicon, informing product advertising decisions.
- Improved coding-task inference throughput by **2.3 times** using **MTP speculative decoding**.
- **Curated evaluation datasets** and simulated **real-world user workloads** to benchmark local LLM inference across models and quantizations.

Apple | *Software Engineer Intern* February 2025 – September 2025

- Improved **occlusion prediction model accuracy** for **AirPods Pro 3** from **95% to 96.2%**, enhancing hearing aid.
- Deployed web application built with **Next.js** and **MongoDB** on Apple's internal **Kubernetes (K8S)** platform and integrated secure authentication with **AppleConnect (OAuth2.0)**; passed **internal security audit (GateHouse)**.
- Presented the project to two Directors and one Senior Director, secured **approval for long-term adoption**.

RBC | *Machine Learning Engineer Intern* September 2023 – April 2024

- Fine-tuned **LLaMA2-70B** with **QLoRA**, adapting it to communicate in the style of a financial analyst, while efficiently operating under constrained computational resources (**A6000**).
- Collaborated on building a **web application**, progressing from **POC to pilot** before transition to the production team. Responsible for implementing backend using **FastAPI** and database with **Postgres**.

Research Experience

Stanford University | *Research Assistant* January 2026 – Present

- Replaced $200 per experiment LLM as a judge API cost with a local **2.2M-parameter neural scorer**, matching LLM-scored accuracy on **55K rationales**.
- Served **Qwen** models on **NVIDIA DGX Spark** via **vLLM** with **AWQ quantization** and **speculative decoding**, and built a health-checking load balancer across model servers.

University of Toronto | *Research Assistant* May 2024 – December 2024

- Built a diffusion model that predicts indoor Wi-Fi/wireless signal strength from building layouts, cutting prediction error by over **50%**.

University of Toronto | *Research Assistant* May 2023 – August 2023

- Built an LLM-powered chatbot that recommends items by searching user reviews, improving recommendation ranking accuracy by **9.16%** over the previous best method.

Publications

- *A Diffusion-Based Propagation Model for Path Loss Prediction in Indoor Environments*, **EuCAP 2025**, Second Author, **7 citations**.
- *Retrieval-Augmented Conversational Recommendation with Prompt-based Semi-Structured Natural Language State Tracking*, **SIGIR 2024**, Co-First Author, **50 citations**.

Open Source Projects

GemmaLLM Link

- Built a **200M-parameter multimodal LLM** in **PyTorch**, pre-trained on **6.5 million English Wikipedia articles** (**2B tokens**) and 2 million images for multimodal data.
- Increased inference throughput by **12.65**× through **KV caching**, **MTP speculative decoding**, and **cuBLAS** on coding tasks.

Leadership Experience

ML Project Director, U of T Machine Intelligence Student Team September 2023 – April 2025

- Led a team to design and implement a **highly scalable data scraping pipeline**, using the collected data to **fine-tune large language models**, including Llama2, Phi-2, and Gemma.

Skills

- **Machine Learning:** PyTorch, LLM SFT Fine-Tuning, GRPO RL training, Data Preprocessing, Transformers
- **DevOps & Database:** Kubernetes, AWS, Google Cloud, Docker, VectorDB, PostgresSQL, MongoDB, vLLM
- **Web Development:** Python, TypeScript, Next.js, FastAPI, Git, OAuth2.0, React